RÉPUBLIQUE DU CAMEROUN

Paix-Travail-Patrie

....

MINISTÈRE DE L'ENSEIGNEMENT SUPÉRIEUR

....

COMMISSION NATIONALE D'ORGANISATION DE L'EXAMEN HIGHER NATIONAL DIPLOMA EXAM (HND)

....

REPUBLIC OF CAMEROON

Peace-Work-Fatherland

....

MINISTRY OF HIGHER EDUCATION

....

NATIONAL COMMISSION FOR THE ORGANISATION OF HIGHER NATIONAL DIPLOMA EXAM (HND) EXAM

....

National Exam of Higher National Diploma 2024 Session

Specialty/option: NWS-SWE-EDM.
Paper: Discrete Mathematics
Duration: 4 hours

Credits: 4 /4/10

ANSWER ALL QUESTIONS

Instructions To Candidates:

Answer all questions. You are authorized to use only non-programmable calculator.

SECTION A: MCQs (20 MARKS)

One (01) Mark for Each Correct Answer.

1) **The probability of selecting at random a man in a crowd containing 20 men and 33 women is,**
   A. 0.6226
   B. 0.05
   C. 0.3774
   D. 1

2) $\lim_{x\to 0}\left\{\frac{\tan x - x}{x^3}\right\}$ **is equal to**
   A. 3
   B. 1/3
   C. 2/3
   D. 0

3) **The Laplace transform** $\mathcal{L}\{2e^{7x-2}\}$ **is equal to**
   A. $\frac{2e^{-2}}{s-7}$
   B. $\frac{2e^{2}}{s-7}$
   C. $\frac{e^{-2}}{2(s-7)}$
   D. $\frac{e^{2}}{2(s-7)}$

4) **The general term of the series $1+\frac{3}{2}+\frac{5}{2^2}+\frac{7}{2^3}+\cdots$, is**

A. $\frac{2n+1}{2^{n-1}}$
B. $\frac{2n-1}{2^{n+1}}$
C. $\frac{2n-1}{2^{n-1}}$
D. $\frac{2n+1}{2^{n+1}}$

5) **When the $\lim_{x\to\pm\infty} f(x) = \pm\infty$, it implies Possibility of**

A. a vertical asymptote
B. a horizontal asymptote
C. a quadratic asymptote
D. an oblique asymptote

6) **The Domain of definition of the function: $f(x) = \frac{x^2-1+\ln x}{e^x-e}$, is**

A. $]0,1[$ u $]1,+\infty[$
B. $]1,+\infty[$
C. $]0,e[$ u $]e,+\infty[$
D. $]e,1[$ u $]1,+\infty[$

7) **The parity of the function $f(x) = \frac{x^3-x}{|x|+4}$ is,**

A. even
B. odd
C. neither odd nor even
D. positive

8) **The equation, log(x -1)+log(x+1)=2log(x+2) has as solution x=**

A. 4/5
B. -5/4
C. 5/4
D. -4/5

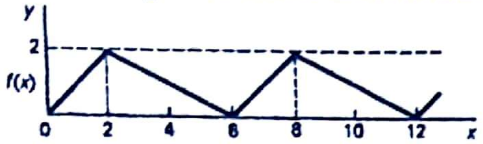
9) **In the Alambert's theorem the $\lim_{n\to\infty}\frac{U_{n+1}}{U_n} =< 1$ *inplies***

A. Convergence
B. Divergence
C. Inconclusive
D. Constant

10) **Three numbers are in arithmetic progression. Their sum is 9 and their product is 20.25. What are the three numbers?**

A. 5/2, 1/2,3/2
B. 7/2,5/2,6/2
C. 3/2,3,9/2
D. 5/2,3,1/2

11) **Determine the period of the function shown graphically below**



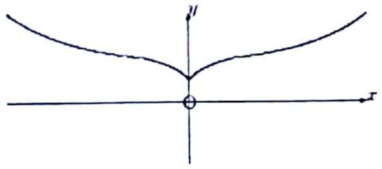
A. 2
B. 8
C. 6
D. 3

12) **The exact solution to the following; $\int_{\frac{1}{2}}^{1} \frac{dx}{x\sqrt{4x^2-1}}$ is**

A. $\pi/2$
B. $\pi/4$
C. $\pi/6$
D. $\pi/3$

$\int^{1}_{1/2}$

13) **The function shown graphically below is**



A. an even function
B. an odd function
C. neither odd nor even function
D. an asymmetric function

14) **Make y a subject in the equation $\ln(1+y) = \frac{1}{2}x^2 + \ln 4$**

A. $y = \frac{1}{2}x^2 + \ln 4 - 1$
B. $y = 4e^{\frac{1}{2}x^2} - 1$
C. $y = 4e^{\frac{1}{2}x^2} - \ln 4$
D. $y = \frac{1}{2}x^2 - 4$

1+. y y=

15) **Find the value(s) of $\frac{dy}{dx}$ of $x^2y + y^2 = 5$ at $y = 1$**

A. $-\frac{2}{3}$ *only*
B. $\frac{2}{3}$ *only*
C. $\pm\frac{2}{3}$
D. $\pm\frac{4}{3}$

16) **When $f(t+T) = f(t)$, T is called**

A. the fundamental period of $f(t)$
B. the period of $f(t)$
C. constant of $f(t)$
D. increment of $f(t)$

17) **which theorem is verified if $f(x)$ is continuous on [a,b], derivable on]a, b[,$f(a) = f(b)$** and there exist at least one number c within]a, b[, such that $f'(c) = 0$,

A. cauchys theorem
B. green's theorem
C. Rolle's theorem
D. Mean value theorem

18) **The following equation,$\oint Pdx + Qdy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dxdy$, expresses the;**

A. Cauchy's theorem
B. green's theorem
C. Rolle's theorem
D. Mean value theorem

19) **$tanh^{-1}\left(\frac{x^2-1}{x^2+1}\right)$ is identical to**

A. $tan\left(\frac{x^2-1}{x^2+1}\right)$
B. $tan^{-1}\left(\frac{x^2-1}{x^2+1}\right)$
C. lnx
D. $\ln(x^2-1)$- $\ln(x^2+1)$

20) **If $f(x) = x^3 - 3^x$, then $f'(x) =$;**

A. $x^2 + ln3x$
B. $3x^2 + ln3$
C. $3x^2 + 3x$
D. $3x^2 + x^3$

3 3 2
3 8 8 5 1 · 7 5
2 0 1 7 2 0
4 9 9 5 5 4
5 2 8 4 4 8
2 5 4 0 4 3
1 4 1 9 6 3 · 25
17 5 4 5 7 9 · 1

21) **The standard deviation from the mean of the set of numbers $\{35; 22; 25; 23; 28; 33; 30\}$ is given by:**

A. 3.50
B. 4.00
C. 4.60
D. 5.50

22) **A box contains 74 brass washers, 86 steel washers and 40 aluminum washers. Three washers are drawn at random from the box without replacement. The probability that all three are steel washers is:**

A. 0.045
B. 0.086
C. 0.0779
D. 0.0668

23) A component is classified as one of top quality, standard quality and substandard with respective probabilities of 0.07, 0.85 and 0.08. the probability that a component is either top quality or standard quality is :

A. 0.85
B. 0.90
C. 0.92
D. 0.96

P(tq) = 0.07
P

24) Machines A, B and C manufacture components. Machine A makes 50% of the components, machine B 30% and machine C the rest. The probability that a component is reliable is 0.93 when made by machine A, 0.90% when made by machine B and 0.95 % when made by machine C. a component is selected at random the probability that it is reliable is :

A. 0.925
B. 0.95
C. 0.90
D. 0.88

P(A) = 0.75
P(B) 0.18
P(C) = 0.15

25) Which of the following is not a solution of the equation: $2 - 4cos^2x = 0$ for x having values in $[0; 2\pi]$

A. 45^0
B. 135^0
C. 225^0
D. 30^0

26) The following equations: $\frac{tanx+secx}{secx(1+\frac{tanx}{secx})}$ is equal to:

A. $cosx$
B. $cos3x$
C. 1
D. $sinx$

27) The gradient of the curve $y = 3x^4 - 2x^2 + 5x - 2$ at the point $(0, -2)$ is

A. 2
B. 4
C. 5
D. 6

28) Let $y = \sqrt[x]{(x-2)}$; if $x = 3$, then

A. $\frac{dy}{dx} = 1$
B. $\frac{dy}{dx} = 2$
C. $\frac{dy}{dx} = 4$
D. $\frac{dy}{dx} = 3$

29) **The average value of the function $f(t) = t + e^t$ over $[0,\ 2]$ is :**

A. 3.2
B. 2.8
C. 4.1945
D. 3.8

30) **The domain of definition of the function $f(x) = \frac{x}{\sqrt{|x+2|}} - \frac{x}{\sqrt{|x+1|}}$ is :**

A. $]-\infty;\ -2[U]-2;\ -1[U]-2;+\infty[$
B. $]-2;\ -1[$
C. $]-2;\ -1[U[-1;\ +\infty[$
D. $]-\infty;\ -2]U]-2;+\infty[$

31) **For the graph of an odd function:**

A. It is Symmetric with respect to the origin
B. Its average value is zero
C. It has a cosine Fourier series
D. Its r.m.s value is zero

32) **The exact value of $\lim_{x \to +\infty} x - \sqrt{x^2 - 2x}$ is :**

A. 1
B. 2
C. .0
D. .$\frac{2}{3}$

33) **A quadratic approximation of $y = x^3$ near $x = 2$ is given by:**

A. $6x^2 - 10x + 8$
B. $6x^2 - 12x + 8$
C. $x^2 - 12x + 6$
D. $6x^2 - 12x - 7$

34) **The value of $\int_0^1 \frac{4x^3-2x^2+3x-1}{2x^2+1} dx$**

A. 0.4
B. 0.6
C. 0.275
D. 0.8

35) **Laplace transform of : $f(t) = 2 - t^2 + 2t^4$ is**

A. $\frac{2}{s} - \frac{2}{s^3} + \frac{48}{s^5}$
B. $2s - 2s^3 + 48s^5$
C. $2s - 4s^3 + 96s^5$
D. $\frac{2}{s} - \frac{4}{s^3} + \frac{96}{s^5}$

36) Find $L^{-1}(\frac{4s-1}{s^2-s})$

A. $1+e^{-t}$

B. $1-3e^{-t}$

C. $1+3e^{t}$

D. $1-e^{-t}$

37) What are the first five terms of the sequence defined as :

$$\begin{cases} a_1 = 3 \\ a_{n+1} = a_n - 4, for\ N \geq 1 \end{cases}$$

A. -3, -2, -1, 0, 1

B. -1, -5, -9, -13, -17

C. 3, -1, -5, -9, -13

D. 3, -1, 0, 1, 2

38) Given that $x \sim P_o(4)$, $var(x) =$

A. 2

B. 4

C. 16

D. 1

Question 39 and 30

In an experiment to determine the relationship between current flowing in an electric circuit and voltage applied, results are obtained in the table below.

Current (mA) (x)	5	11	15	19	24	28	33
Voltage (y) applied (V)	2	4	6	8	10	12	14

39) The correlation coefficient between these results is :

A. 0.77

B. 0.88

C. 0.99

D. 0. 66

40) The equation of the regression line of applied voltage Y on current X is given by:

A. $Y = 1.02X + 2.4$

B. $Y = 1.02X - 2.4$

C. $Y = 1.142 + 2.268X$

D. $Y = 1.142 - 2.268X$

SECTION B:(60 MARKS)

I. Analysis (30 Marks)

1. Find the Fourier coefficients of the periodic function $f(x)$ given by
$f(x) = \begin{cases} -k \text{ if } -\pi < x < 0 \\ k \text{ if } 0 < x < \pi \end{cases}$ and $f(x + 2\pi) = f(x)$ **(6 marks)**

2. Find the Laplace transformation of the function $f(t) = e^{3t}sinh(t)$ and find the original function $f(t)$ if its Laplace transform is $F(s) = \frac{5s+1}{s^2-25}$. **(6 marks)**

3. Consider a function of two variables $f(x, y)$ and df the total differential of f. The total differential is said to be an exact differential if $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$ when there exists. If $df = \left(2xy^3 + \frac{2}{y}\right) dx + \left(3x^2y^2 - \frac{2x}{y^2}\right) dy$ show that df is an exact differential. **(5 marks)**

4. Verify Green's theorem in the plane for: $\oint_C (2xy - x^2)dx + (x + y^2)dy$ Where C is the closed curve of the region bounded by $y = x^2$ and $y^2 = x$. **(5 marks)**

5. Solve the homogeneous equation.
$(x^2 + y^2)dy = xydx$, given that $x = 1$ when $y = 1$ **(4 marks)**

6. Determine the general solution of the differential equation:

$$\frac{d^2y}{dx^2} - 3\frac{dy}{dx} - 4y = 3e^{-x}$$ **(4 marks)**

II. Statistics (30 Marks)

1. The following table summarizes the masses, measured to the nearest microgram(μg), of 200 microchips of the type.

Mass (μg)	*Frequency*
70 – 79	7
80 – 84	30
85 – 89	66
90 – 94	57
95 – 99	27
100 – 109	13

(i) Calculate estimates of the median and upper quartile of the distribution.
(ii) Estimate the number of microchips whose actual masses are less than 81 μg.
(iii) Calculate estimates of the mass and the standard deviation of the distribution.

(5+3+5 = 13 marks)

2. The marks obtained by 8 SWE students in Discrete mathematics and Digital Electronics at a given HND examination are tabulated below

Student	A	B	C	D	E	F	G	H
Maths Marks (x)	45	23	27	33	18	0	8	14
D. Electronics marks (y)	31	20	18	33	19	1	13	9

a) Find the equation of the least square regression line y on x
b) Calculate the product moment correlation coefficient
c) Calculate the minimum sum of squares of residuals of y on x
d) Predict the Digital Electronic score of a student who scores 30 marks in Discrete maths
e) Calculate Kendall's coefficient of rank correlation between Discrete maths and Digital Electronic

(5+3+ 3+2+ 4= 17 marks)

III. Probability (20 Marks)

1. A discrete random variable X has probability mass function defined by

$$f(x) = \begin{cases} \dfrac{k(x-1)}{6}, for\ x = 2,3,4,5,6,7 \\ \dfrac{13-x}{36}, for\ x = 8,9,10,11,12. \end{cases}$$

Calculate

a) The value of the constant k **(2 marks)**
b) $E(X)$ and $Var(X)$ **(5 marks)**
c) $p(6 \le X \le 9)$

Another random variable Y is connected to X by the relationship $Y = (5X + 3)$. Calcualte

d) $E(X + Y)$ **(3 marks)**
e) $Var(X + Y)$ **(2 marks)**

2. A batch of 40 components contains 5 which are defective. A component is drawn at random from the batch and tested, then a second component is drawn. Determine the probability that neither of the components is defective when drawn

(a) With replacement, and **(4 marks)**

(b) Without replacement. **(4 marks)**